

LUKE TAYLOR

GAMES PROGRAMMER

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Personal Statement

Games Programmer and recent graduate from Staffordshire University's Computer Games Programming course, with a strong foundation in C++, C#, object-oriented development, and low-level performance optimization. Recent work includes: reducing physics update time from 250ms to 1.9ms over 8,000 colliding objects; memory pooling and infrastructure work on a custom 2D game engine; using docker to build and cross compile a C++/SDL game to a custom ARM linux console; and a procedural dungeon generation tool for Unity integrated with their TileMaps system. I'm looking for a junior role where I can further build my skillset alongside an experienced team.

Projects

Sparrow Engine (Collaborative, C++, OpenGL, 2026)

- Team effort to create a game engine capable of deploying a game with physics, audio and input
- Built infrastructure including memory pooling and time handling across the engine
- Created a factory system using ImGui to build UI windows and features easily in the codebase
- Built tools for editor such as profiling and asset browser windows
- Collaborated via Git using feature branches and PR reviews, maintaining consistent code style and documentation

Procedural Dungeon Generator (Solo, C#, Unity, 2026)

- Designed and created an agent based PCG algorithm capable of generating random, error-free layouts with given parameters
- Prototyped an interior generation system that uses a restriction based approach to fill rooms
- Integrated with Unity Tile-Maps feature for easy customization in editor using exposed parameters

Low Level Optimization (Solo, C++, OpenGL, 2025)

- Benchmarked low-level optimisation techniques rendering 8,000 simultaneously colliding objects, profiling with Optick and custom in-engine instrumentation
- Built a thread pool with spatial collision partitioning to distribute work across cores and eliminate contention
- Implemented a fixed-size memory allocator with deterministic allocation and built-in tracking/validation
- Reduced physics update time from 250.5ms to 1.9ms

Evercade Console Game (Solo, C++, SDL, Evercade, 2024)

- Wave-based 2D fighter game built with SDL and C++ for the Evercade EXP console
- Used Docker to port to custom ARM Linux system
- Implemented platform-specific controls
- Multiple mechanics and abilities for player and enemies
- Includes full game loop, animations, collision detection, and a progressive wave system

Skills

Programming Languages

C++, C#, Python, Lua

Game Engines

Unity, Unreal Engine, Godot

Frameworks

MonoGame, OpenGL, SDL

Version Control

Git & Github

Other

Procedural Generation, Networking, Low Level Optimization, Docker, Tools Programming, HTML & CSS

Education

University of Staffordshire (2023 - 2026)

BSc Hons Computer Games Programming -
Second Class : 1st Division

Shrewsbury College (2021 - 2023)

DDD - BTEC Level 3 Information Technology

Shrewsbury Academy (2016 - 2021)

9 GCSEs Including Maths, English and Science

Interests

Procedural Content Generation

I have a strong interest in PCG and developing tools and applications that benefit both game design and mechanics.

Game Mechanics

I have a passion for interesting game mechanics and features, I find myself always thinking of new ideas for future projects and taking inspiration from recently released games.

Chess

I've always enjoyed the game of chess, playing against family and friends, even creating my own AI chess bot as a personal project.

Work History

Warehouse Operative, DX

Shrewsbury, July 2022 - July 2023

In-Store Worker, Papa Johns

Shrewsbury, October 2021 - July 2022